

The exact lower drift contractions

HOCF–RIESZ, Round 014

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Status and premises. This synthesis records the THM–035 conditional implication. Complete constructor and independently reconstructed proofs are sealed separately; AUD–047 records their full comparison; separate fresh hostile review AUD–048 also passes. Root verified all three sealed packets and reran their exact diagnostics. The R8 genuine-domain and R12 uniform smaller-gradient premises enter only in their exact stated scopes. No cubic-smallness or fluctuation-law conclusion is inferred here.

An integrable product with a uniform power

Fix $d \geq 3$, $0 < s < d - 2$, $3s < 2d - 2$, finite $T, L, \nu_* \geq 0$ and a smooth real terminal test h . Use the unit Haar torus, coefficient-one periodic Riesz force $K = -\nabla g$, zero external drift, the actual Fourier test and genuine symmetric terminal-zero full pair inverse Φ , with both responses. For $N \geq 2$, take $0 \leq \nu \leq \nu_*$ and $\nu N^{2/p} \leq L$, $p = s + 2$. Define

$$q_- = \max(1, s/2), \quad q_+ = \min(d/2, d - s - 1, s + 1), \quad q = (q_- + q_+)/2.$$

Every upper endpoint strictly exceeds both lower endpoints, so

$$1 < q < d/2, \quad q < s + 1, \quad s + 1 + q < d, \quad q > s/2.$$

With the fixed periodic weights $w_a = w_a^q$, equal to $|z|^{-a}$ near zero, the admitted R12 estimate is

$$|\nabla_{x,y} \Phi_t(x, y)| \leq A_q N^{\alpha_q} w_q(x - y), \quad \alpha_q = (s + 1 - q)/p,$$

where A_q is independent of N, ν, t in this fixed family. Set

$$B\Phi = K(x - y) \cdot (\nabla_x - \nabla_y)\Phi, \quad g_t(x) = \int B\Phi_t(x, y) dy, \quad c_t = \int g_t(x) dx.$$

The product norm gives

$$|B\Phi_t(x, y)| \leq \sqrt{2} A_q N^{\alpha_q} |K(x - y)| w_q(x - y).$$

Its relative singular exponent is $s + 1 + q < d$, so

$$J_q = \int |K(z)| w_q(z) dz < \infty, \quad \sup_t \|g_t\|_\infty, \quad \sup_t |c_t| \leq D_q N^{\alpha_q}, \quad D_q = \sqrt{2} A_q J_q.$$

For an explicit bound, on a fixed embedded ball of radius $R = 1/16$ write $K(z) = sz|z|^{-s-2} - \nabla H_s(z)$. Then

$$J_q \leq \frac{s|\mathbb{S}^{d-1}|R^{d-s-1-q}}{d-s-1-q} + \frac{\sup_{B_R} |\nabla H_s| |\mathbb{S}^{d-1}|R^{d-q}}{d-q} + \sup_{z \notin B_R} |K(z)| w_q(z).$$

All denominators are positive. Moving-tube dominated convergence identifies continuous genuine R8 contractions at every remaining coordinate. Absolute Fubini gives $c_t = \int g_t$; no diagonal trace is assigned.

Both exact contractions and the pathwise bound

For the actual empirical measure η_N and $\rho_N = \eta_N - dx$, the lower-plus-scalar drift is exactly

$$\ell_N(t) = \frac{1}{N} \rho_N[g_t] + \frac{1}{2N} c_t = \frac{1}{N} \eta_N[g_t] - \frac{1}{2N} c_t.$$

Both terms remain. Every empirical probability measure satisfies

$$|\ell_N(t)| \leq \frac{3}{2} D_q N^{\alpha_q - 1}.$$

For positive noise put $b = \min(1, \nu^{-1})$, $\sigma = \sqrt{Nb}$; at zero noise set $b = 1$, $\sigma = \sqrt{N}$ directly. Consequently, pathwise,

$$\sigma \int_0^T |\ell_N(t)| dt \leq \frac{3}{2} T D_q \sqrt{b} N^{-\kappa_q}, \quad \kappa_q = \frac{2q - s}{2(s + 2)} > 0.$$

Taking expectation costs no law-comparison factor. The actual iid-Haar-prepared singular process meets the earlier domain premises; positive-time independence is not assumed. Zero time and zero noise are included directly.

For clarity, let $D_2[F] = N^{-2} \sum_{i \neq j} F(X_i, X_j)$ and use the exact ordered statistic $P_N = U_2/2$. Splitting the force-generator sum at its repeated label gives $D_2[B\Phi]/(2N)$ and the three-distinct contribution. Integrating the six-permutation average

$$C\Phi = \text{Sym}_3[K(x - z) \cdot \nabla_x \Phi(x, y)]$$

retains both response slots with coefficient one. Subtracting the inverse's internal operator term yields precisely

$$\frac{1}{2N} D_2[B\Phi] - \frac{1}{N} P_N[B\Phi] = \frac{1}{N} \rho_N[g_t] + \frac{1}{2N} c_t.$$

The full conditional R8 identity is therefore

$$dP_N[\Phi_t] = \{-P_N[J_t] + U_3[C\Phi_t] + \ell_N(t)\} dt + dM_t^2.$$

Here U_3 has denominator N^3 and all three Haar-background levels. At $N = 2$ only its three-distinct particle sum is empty; its backgrounds remain. The constructor's malformed font fragment for “distinct” has a separate rendering erratum, preserving every sealed source byte and coefficient.

Critical inclusion and the remaining assertion

If $s < d - 2$ and $s(s + 2) < 2d$, then $3s < 2d - 2$: for $s \leq 2$ use $2d > 2s + 4 \geq 3s + 2$; for $s > 2$ use $s(s + 2) - 2 = 3s + (s - 2)(s + 1) > 3s$. For $\lambda_N = \beta_N N^{s/d-1} \rightarrow \lambda \in (0, \infty)$,

$$\nu_N N^{2/p} = \lambda_N^{-1} N^{s(s+2-d)/(d(s+2))} \rightarrow 0.$$

Thus each such sequence eventually lies in a fixed bounded-rescaled family. Writing $A_N = \sigma_N \int_0^T U_3[C\Phi_t] dt$ and $Z_N = A_N + \sigma_N \int_0^T \ell_N(t) dt$, reverse triangle gives

$$|\mathbb{E}|Z_N| - \mathbb{E}|A_N|| \leq C \sqrt{b_N} N^{-\kappa_q} \rightarrow 0.$$

This is the exact equivalence of the two scaled integrated L^1 targets. It proves neither target small. In particular $\mathbb{E}|\int U_3 dt|$ and $\mathbb{E}\int |U_3| dt$ are not interchanged. Old energy-floor, full subcritical and finite positive critical conditions remain distinct.